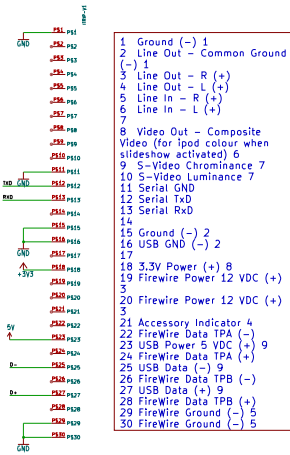


Edge connector into iTrip

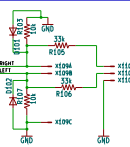


Generic Jumpers



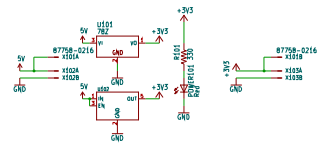
Landing for wires with no other connections

Utility Voltage dividers



Can be used for CV in (0-12V, 0-5V), or to scale 3.3V out to 1V, or ...

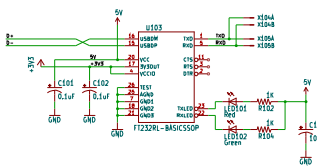
Voltage Regulators



A T0-92 through hole footprint and a SOT-23-5 SMT footprint are provided in parallel, generally use only U2, U3, or U1. U1 has a builtin 5V to 3.3V regulator, 3V3OUT1.

5V can come from USB via the iTrip connector, or 3.6V (2xAA), 4.2V, or 6V+ batteries, etc. as long as the regulators are right sized. 3.6V and lower battery packs can be wired directly to the 3.3V connectors, and no regulator is needed. There's no power mixing on the 3.3V line, so it's not generally safe to charge batteries on 3.3V from the regulators, though expert usage allows both the regulator hardware to coexist with 3.3V batteries. The FTDI 3.3V regulator output is a bit too low to reliably charge batteries.

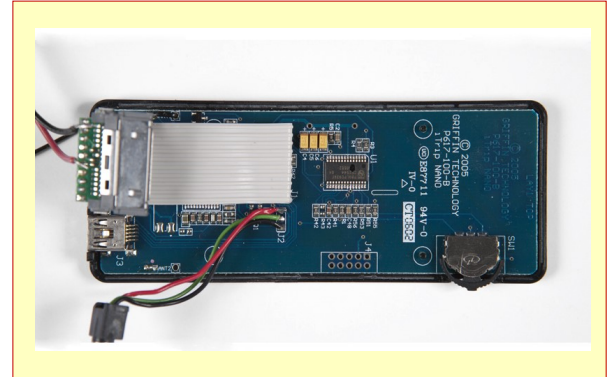
Communication



The FTDI232RL is not the cheapest chip, but it doesn't require any external crystal, EEPROM, or power conversion.

The USB connector on the iTrip is routed to the USB to serial converter, and the UART outputs are run back to the LPC9356 8051 microcontroller inside the iTrip. A single FTDI chip can power and communicate with up to 8 iTrips, but only after configuring higher TX pin drive strength via the FT_PROG software. Each iTrip must be configured during firmware programming to a unique MIDI channel, which also limits the number of connections to 8 (16 channels per MIDI bus, and each iTrip responds to 2 channels). Wire the TX of other iTrips to TX of this board, RX to RX. See detailed instructions in the github readme.

iTrip Internals

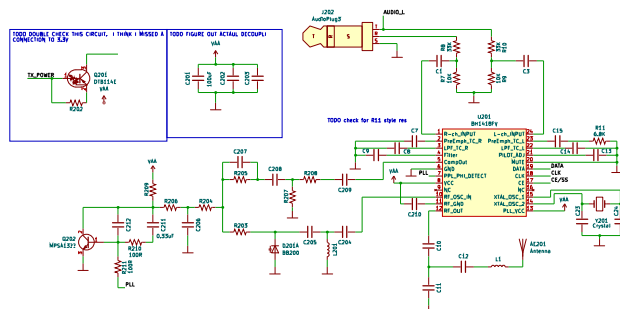
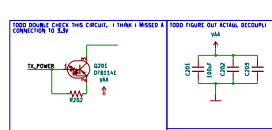
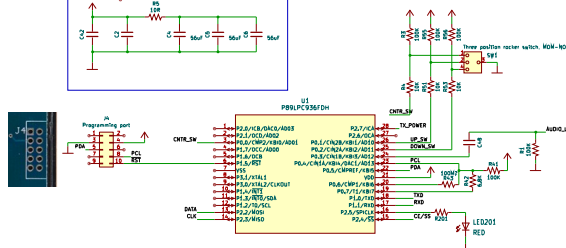
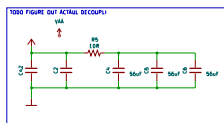
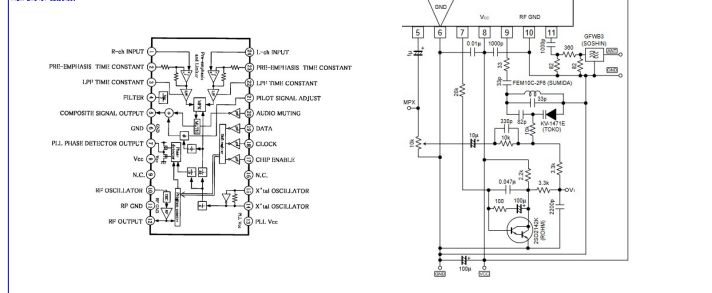


from <http://www.kin-circuits.com>

REVISION HISTORY				
ZONE	REV	DESCRIPTION	DATE	APPROVED
	5	Edge Input		
	6	Tapout PCB and add lots of documentation to schematics		

Responsible dept.	Technical reference	Created by	Approved by	
		Document type	Document status	
		Title, Supplementary title		
		Exitrip Adaptor		
		Rev. Date of issue	Lang. Sheet	
		6 2024-05-18	1/2	

1 Ground (-) 1
 2 Line Out - Common Ground (-)
 3 Line Out - R (+)
 4 Line Out - L (+)
 5 Line In - R (+)
 6 Line In - L (+)
 7 Video Out - Composite Video (for ipod colour when slideshow activated) 6
 8 S-Video Chrominance 7
 9 S-Video Luminance 7
 10 Serial GND
 11 Serial Ix0
 12 Serial Rx0
 13 Ground (-) 2
 14 USB GND (-) 2
 15 3.3V Power (+) 8
 16 Firewire Power 12 VDC (+) 3
 17 Firewire Power 20 VDC (+) 3
 18 Accessory Indicator 4
 19 Firewire Data TPA (-) 9
 20 USB Power 5 VDC (+) 9
 21 Firewire Data TPA (+) 9
 22 USB Data (-) 9
 23 Firewire Data TPB (-) 9
 24 USB Data (+) 9
 25 Firewire Data TPB (+) 9
 26 Firewire Ground (-) 5
 27 Firewire Ground (-) 5



REVISION HISTORY				
ZONE	REV	DESCRIPTION	DATE	APPROVED

Responsible dept.	Technical reference	Created by	Approved by	
	Kin Circuits	Document type	Document status	
		Title, Supplementary title		
		Exitrip Adaptor	Rev. Date of issue 6 2024-05-18	Lang. Sheet 2/2